Branching Fidelity

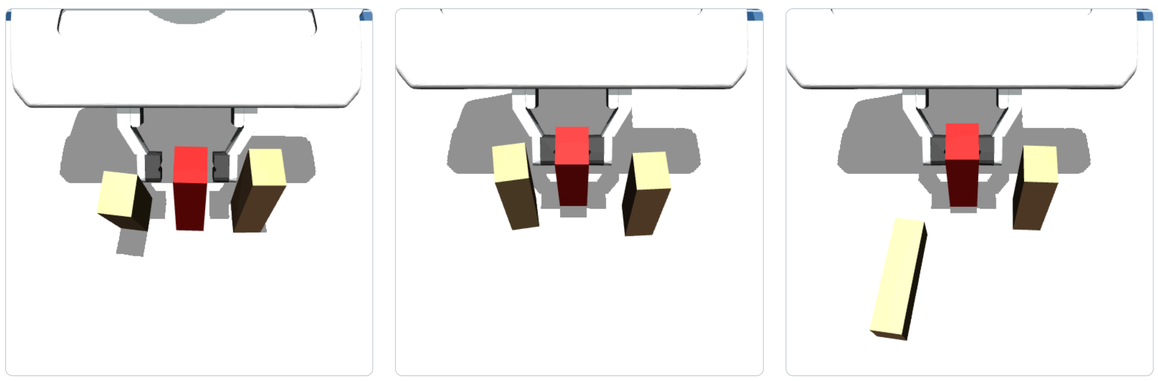
Testing World Models as Intuitive Physics Engines for Robot Safety

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Where a robot's own small errors decide the outcome, its futures fork. A fork marks the edge of the safe set, and a world model is only useful there if it forks where physics does.

1 The question

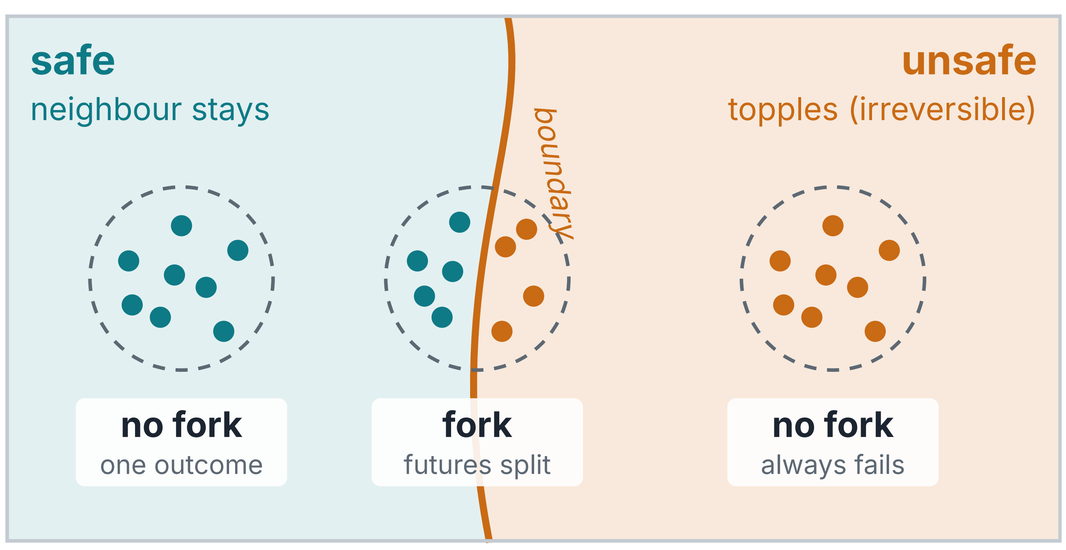
Before each action chunk, a robot should ask: **could my own small errors change what happens?** The same command can lift the middle block cleanly or tip a neighbour, depending on a few millimetres of error.



the chunk starts | half the imagined futures: neighbour stays | the other half: neighbour topples

2 A fork is the edge of the safe set

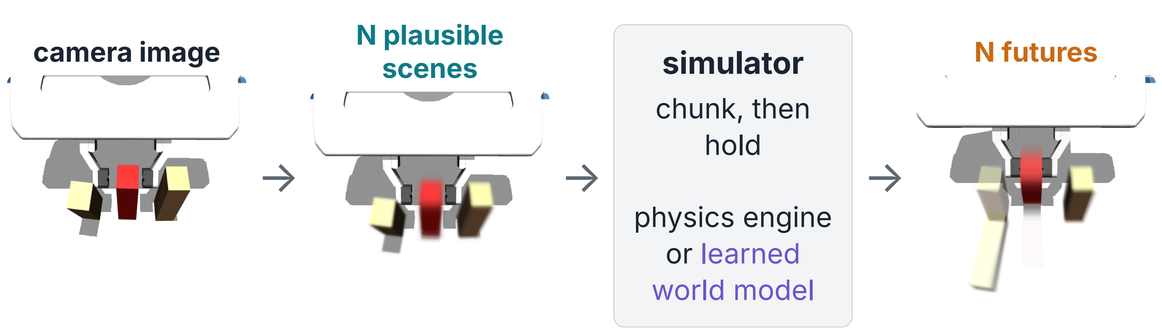
A state is a **fork** if futures sampled from the robot's own uncertainty split into different outcomes. When one branch is irreversible (a toppled block stays toppled), a fork marks the **boundary between safe and unsafe states**, with no failure labels.



A proxy, not a failure detector: a fork with two harmless branches is a false alarm, and an action that always fails never splits.

3 The fork probe

People judge whether a stack will fall by **imagining it a few times** with slightly different guesses of where things are; if the outcomes disagree, it is precarious (the **intuitive physics engine** of cognitive science). The probe does the same with the robot's measured errors.



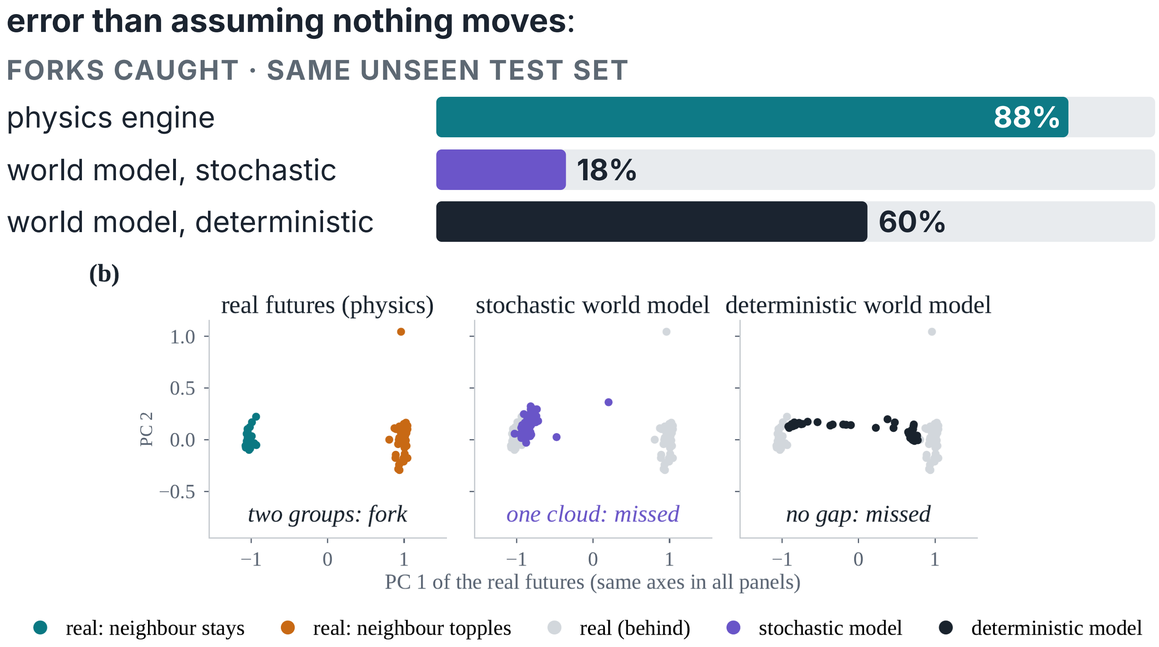
- Samples the robot's **measured** perception and execution errors.
- Flags **two separate groups**, not just spread: futures can vary smoothly with one outcome; a fork has a gap.
- No task labels, failure data or tuned thresholds.

Physics engine = upper reference. MuJoCo with the true masses and friction is not deployable, but sets the bar: **88%** of forks caught, **7%** false alarms, unseen layouts, one camera image.

4 Forks test a world model's physics

To find forks, a simulator must respond to **millimetre changes in state and action** as physics does. Average error, dominated by moments where nothing dramatic happens, does not test this: we call it **branching fidelity**.

World models predicting in DINOv2 feature space, both with **3–4× lower error than assuming nothing moves**:



5 Why

In physics, futures differ only because starts and commands differ, and diverge only near the boundary.

Stochastic: noise in the wrong place
Fresh noise at every step: the right *amount* of spread, but in every direction, almost as much on safe states as on forks (1.3× vs 4.8× in physics).

Deterministic: right place, no gap
Spread comes only from perturbed inputs, as in physics: along the branch, 32× larger on forks. But squared-error training predicts the average, blurring the branches.

6 What closes the gap

Put the uncertainty where it is, in the robot's knowledge of the scene: perturb the start, roll out deterministically. **Train on the forks**: the split test finds each training scene's rare branch without labels; we oversample it.



False alarms 7–11%. A generative model given the same uncertain starts still catches 0%: noise in the dynamics erases the forks.

7 Takeaways

Benchmark branching fidelity
Beside average error: does the model fork where physics does?

Uncertainty in the start
Deterministic dynamics from perturbed starts keep the branch; per-step noise smears it.

Train for rare branches
Oversampling them, found without labels, lifts catch rates.

Pair forks with harm
A fork is a proxy: add a harm signal, e.g. can the outcome be undone? Our label-free safety filter builds on this.

Limitations: simulation, one three-block task; the ground truth uses the same error model the probe samples; small world models, trained once.